

ÁNGEL ESTEBAN FÉLIZ FERRERAS

Applied Scientist | Sequential Decision Analytics & Reproducible Optimization

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PROFESSIONAL SUMMARY

Applied Scientist specialized in **Sequential Decision Analytics** (Warren B. Powell framework). I design and deploy **online decision policies under uncertainty** that deliver measurable economic impact.

Mastered two of Powell's four policy classes (**PFA** and **DLA**) and built production-grade simulation environments (Monte Carlo & Discrete Event). Expert in turning massive observational datasets into optimal real-time decisions using R stacks, DuckDB, tidymodels, and fully reproducible MLOps (Nix + Docker + GitHub Actions).

Proven ability to prevent **\$500K+ annual revenue leakage** and generate **51.5% uplift** in expected earnings through simulation-driven policies — exactly the skill set required for roles in stochastic optimization, revenue management, dynamic pricing, and high-uncertainty operations.

TECHNICAL SKILLS

Sequential Decision Analytics

- Powell framework (PFA & DLA implemented)
- Lookahead simulation, opportunity-cost labeling, policy evaluation
- Monte Carlo & Discrete Event Simulation

Core Stack

- R: data.table, DuckDB (60 GB+), tidymodels (XGBoost), targets, pins
- Python: Flask, SQLAlchemy, pandas
- Reproducibility & MLOps: Nix, Docker/Docker-Compose, GitHub Actions, roxygen2/testthat
- Data Engineering: spatial joins (OSM + ACS), web scraping, automated validation (tidyvalidate)

Other

- Full-stack MVC, Quarto reporting, high-performance parallel processing

RELEVANT PROJECTS & OPEN SOURCE

[NYC Taxi Driver Earnings Optimizer](#) (Sequential Decision Analytics)

- Built end-to-end **DLA policy** (15-minute lookahead simulation with dynamic radius) to create ground-truth target take_current_trip from 67 GB raw TLC data.
- Engineered **PFA** (XGBoost + optimized threshold 0.22) delivering **\$0.35 net benefit per trip** (+51.5% vs. "accept-all" baseline).
- Developed forensic infrastructure with DuckDB + custom tidymodels recipes; spatial enrichment with OSM & ACS; full Nix/Docker reproducibility.

[Driver Hypertension Risk Pipeline](#) (Reproducible Statistical Analysis)

- Built a fully reproducible R data pipeline using {targets} to process raw survey data from public transport drivers and identify hypertension risk factors (t-tests, ANOVA, hierarchical clustering with Ward's method).

- Performed end-to-end feature engineering (BMI categorization, dietary habit clustering) and statistical inference on blood pressure, stress, and lifestyle variables.
- Ensured 100% reproducibility and automation with Nix + Docker, remote data access via {pins} (board_url()), and GitHub Actions that automatically run the pipeline and deploy the interactive Quarto HTML presentation to GitHub Pages on every push.

Open-Source R Packages

- [{tidyvalidate}](#) – Production-grade validation layer on top of {validate} with automatic error/warning handling and row-level reporting.
- [{biblegatewayr}](#) – Web-scraping package for programmatic Bible verse retrieval.

[CineFlow](#) – Full-Stack Ticket Management System

- Developed complete MVC Flask + SQL Server application (user, admin, gatekeeper roles) with Nix/Docker reproducibility and SSH development access.

PROFESSIONAL EXPERIENCE

Phoenix Tower International (PTI) – Leasing Reporting Manager

April 2021 – Present | Remote (17 countries)

- Designed and deployed automated R QA system enforcing 42 complex billing rules → prevented **\$500K+ annual revenue leakage**.
- Created internal R package {KpiReport} (unit-tested, documented) and open-sourced [{ExcelSummary}](#) + [{tidyvalidate}](#) to eliminate manual errors.
- Led data migration from Siterra to Sitetracker while preserving \$15 M+ revenue logic and compliance rules.

EDUCATION & CERTIFICATIONS

- **Master of Data Science** – Spain Business School (expected Sep 2026)
- **B.S. Industrial Engineering** – INTEC (GPA 3.91/4.0, 2021)
- **DataCamp Certifications:** [Associate Data Scientist in R](#), [R Programmer](#), [Web Scraping in R](#)